Model Card: Home Credit Default Risk Model

LightGBM credit risk score, version 1.0 · IS 6812 MSBA Capstone

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Executive summary

> **Recommendation: approve a 3-month shadow test, then a controlled pilot, of the risk score at a 10% default-probability cutoff, after removing marital status from the model.**

What the model does. It estimates each applicant's probability of payment difficulty using only information Home Credit already collects at application time. It ranks risk reliably: on applicants it never saw, it puts a defaulter above a repayer 77% of the time (AUC 0.769), confirmed on Kaggle's hidden test set (0.763).

Expected financial impact at the recommended cutoff (back-tested on 307,511 historical loans, researched cost assumptions):

	Current practice (approve all)	Model cutoff 10%
Approval rate	100%	75%
Default rate among approved loans	8.1%	4.2%
Net profit (billions, local currency)	3.94	6.79

- Net profit rises by about **2.9 billion (+72%)**, because the defaults avoided cost far more than the margin given up on good borrowers who are declined.
- The default rate among approved loans falls from 8.1% to 4.2%, well beyond the project goal of a 10% relative reduction.
- **The trade-off:** about 25% of applicants would be declined. If Home Credit's real loss on a default is closer to 45% (the Basel benchmark), the best cutoff rises to about 15% and approvals to about 85% (Section 4).

Key caveats

1. **Cost assumptions drive the cutoff.** Margin and recovery come from published sources, not Home Credit's internal books; the cutoff must be reset with actual figures.
2. **Compliance fixes before any live use.** The model uses marital status, which U.S. fair lending law (ECOA / Regulation B) prohibits, and age and region, which need legal review. Gender is excluded.
3. **Fairness gaps exist.** Men and applicants without higher education are approved less often. The gaps track real repayment differences but must be reviewed formally (Section 7).
4. **Thin-file applicants are scored less accurately**, which matters for a lender whose mission is serving them (Section 8).

1 Model details

	Detail
Algorithm	LightGBM gradient-boosted decision trees (binary classification)
Version	1.0 (October 2026), LightGBM 4.7.0
Training data	Home Credit Default Risk (Kaggle), application_train.csv: 307,511 loans, default rate 8.07%
Inputs	249 application-time features from data_preparation.py (gender excluded)
Output	Probability of payment difficulty (0 to 1); higher = riskier
Number of trees	2,593 (selected by early stopping on the full data)
Learning rate	0.0103
Leaves per tree	42
Minimum applicants per leaf	100
Rows / features sampled per tree	84% / 38%
L2 regularization (lambda)	1.45
Validation	3-fold stratified cross-validation; Kaggle hidden test set (48,744 applicants)
Owner	Marco Rodriguez, University of Utah MSBA

How it was built. Five candidate models were compared with cross-validation (two logistic regressions, a random forest, XGBoost and LightGBM). LightGBM ranked best and was tuned with a 20-setting randomized search; no resampling was applied, so the predicted probabilities stay calibrated (Section 3.3). Full details are in the modeling notebook.

2 Intended use

Primary users. Credit underwriting and risk teams at Home Credit, and loan officers who review borderline applications.

Decisions it informs.

- Approve or decline a new consumer loan application, using the probability score against a business-set cutoff.
- Route applications near the cutoff to manual review.
- Provide the principal reasons for a decline (adverse action notices, Section 6).

Out of scope.

- Setting interest rates, credit limits or loan amounts (the model was not trained for pricing).
- Collections, fraud detection or marketing targeting.
- Fully automated decisions without human oversight, or use in markets and products unlike the training data (for example mortgages).
- Any decision that uses the score as a proxy for a person’s character or worth.

3 Performance metrics

3.1 Ranking quality (AUC)

	AUC	What it measures
Cross-validated	0.7691	Out-of-fold on 307,511 training applicants (3-fold CV)
Kaggle private	0.7633	Kaggle hidden test set, private leaderboard (~80% of 48,744)
Kaggle public	0.7659	Kaggle hidden test set, public leaderboard (~20%)
Majority class	0.5000	Benchmark: approve everyone (no ranking ability)

AUC answers “how often does the model rate a random defaulter as riskier than a random repayer?” A score of 0.77 is solid for application-only data; the Kaggle scores are within 0.006 of the cross-validated estimate, so the model generalizes to new applicants.

3.2 Precision and recall at the chosen threshold

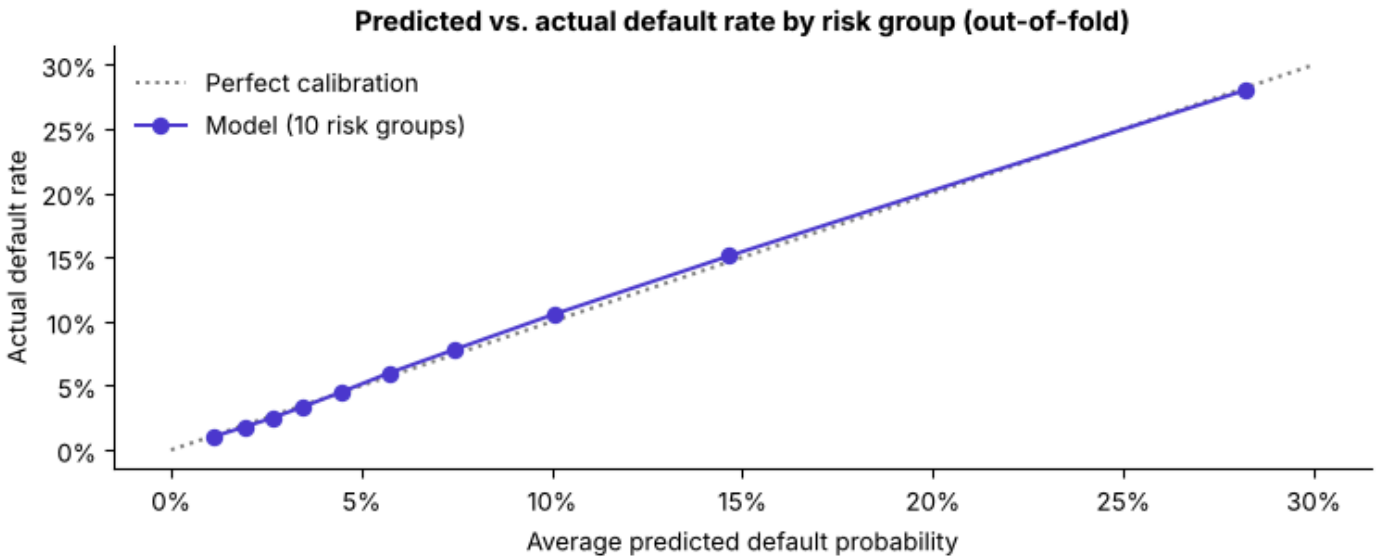
Applicants whose predicted probability is at or above **10%** are flagged as likely defaulters (declined).

	Predicted default (decline)	Predicted repay (approve)
Actually defaulted	15,041	9,784
Actually repaid	61,790	220,896

	Value	Plain-English meaning
Precision	19.6%	Of the applicants declined, the share who would have defaulted
Recall	60.6%	Of all defaulters, the share the model catches
Specificity	78.1%	Of all good borrowers, the share still approved
Accuracy	76.7%	Share of all decisions that match the actual outcome

Precision looks low only because defaults are rare (8%): a declined applicant is still about 2.4 times as likely to default as the average applicant. Recall shows the model stops 61% of defaults while approving 78% of good borrowers.

3.3 Calibration: can the probabilities be taken literally?



The points sit on the diagonal: applicants given a 15% probability default about 15% of the time. That is what makes the cost-based threshold in Section 4 valid.

4 Decision threshold analysis

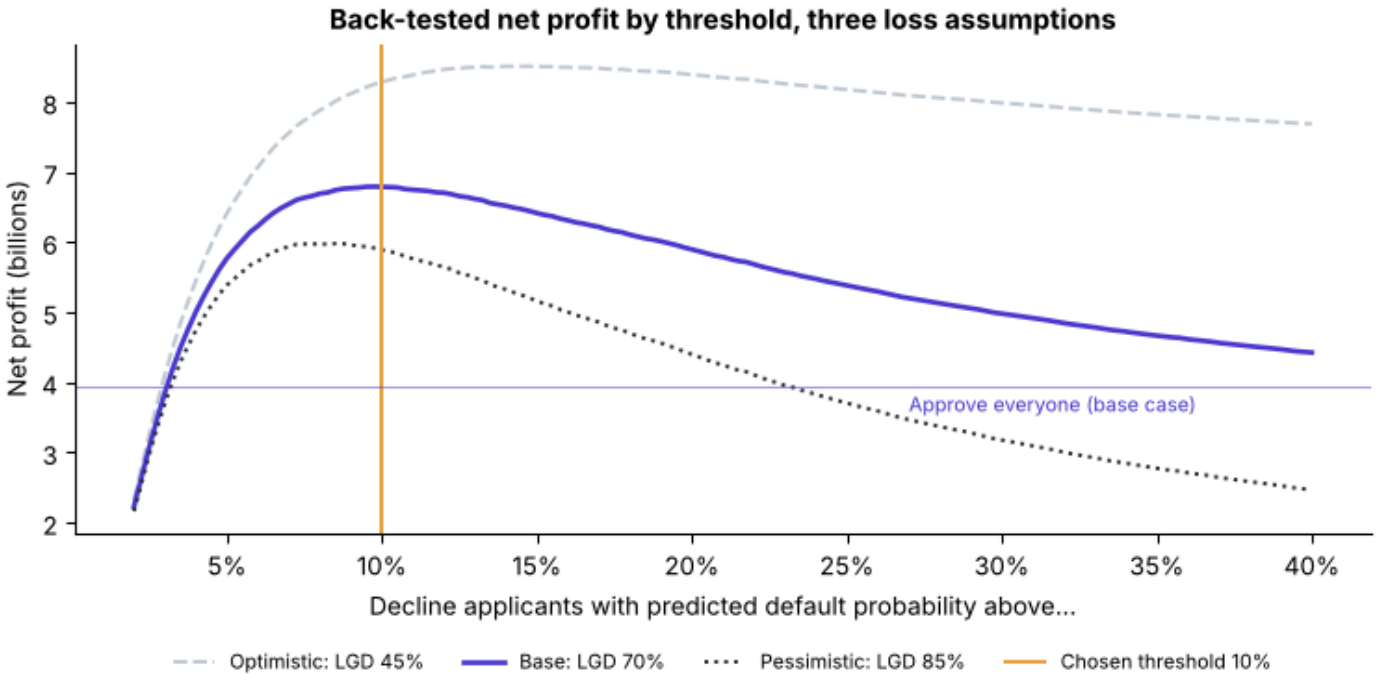
4.1 Cost assumptions and sources

	Value	Basis
Net interest margin	15.8%	Home Credit B.V. FY2018 results [1]
Cost-to-income ratio	42.2%	Home Credit B.V. FY2018 results [1]
Pre-loss margin	9.1% per year	Net interest margin × (1 − cost-to-income)
Typical loan term	20 payments	Median loan amount ÷ installment in the training data
Profit on a repaid loan	8% of loan amount	Annual margin × 52% average balance × term in years
Recovery on a defaulted loan	30%	Net discounted recovery, unsecured consumer loans, European lender [3]
Loss given default (LGD)	70% of loan amount	1 − recovery; Basel benchmark for unsecured claims is 45% [2]

- **Profit on a repaid loan.** Home Credit reported a 15.8% net interest margin and a 42.2% cost-to-income ratio for 2018 [1], leaving about 9.1% a year before credit losses. A typical loan here runs 20 installments and amortizes, so on average about half the principal is outstanding: lifetime profit ≈ **8%** of the amount lent.
- **Loss given default.** Collections research on unsecured consumer loans at a European bank found net discounted recoveries of about 30% for better-performing defaulters and 15% for the worst [3], so LGD ≈ **70%** (range 70 to 85%). The Basel framework assigns 45% to senior unsecured claims [2], used here as the optimistic case. For context, Home Credit's 2018 cost of risk was 9.4% of loans [1].

4.2 Choosing the threshold

Approving an applicant with default probability p is worth $(1 - p) \times \text{margin} - p \times \text{LGD}$. That is positive while $p < \text{margin} \div (\text{margin} + \text{LGD}) = \mathbf{10.3\%}$. Because the model is calibrated, this break-even point should match the back-tested optimum, and it does: the profit curve below peaks at 9.8%. The operating threshold is set at **10%**.



4.3 Sensitivity analysis

Outcomes at different thresholds (base assumptions):

Threshold	Approval rate	Default rate (approved)	Precision	Recall	Net profit (bn)
5%	49.7%	2.6%	13.5%	83.8%	5.77
8%	67.7%	3.7%	17.2%	69.1%	6.69
10%	**75.0%**	**4.2%**	**19.6%**	**60.6%**	**6.79**
12%	80.4%	4.7%	21.8%	52.9%	6.71
15%	85.9%	5.3%	24.7%	43.3%	6.42
20%	91.5%	6.1%	29.6%	31.1%	5.90
30%	96.9%	7.1%	38.7%	15.0%	4.98
Approve all	100.0%	8.1%	n/a	0.0%	3.94

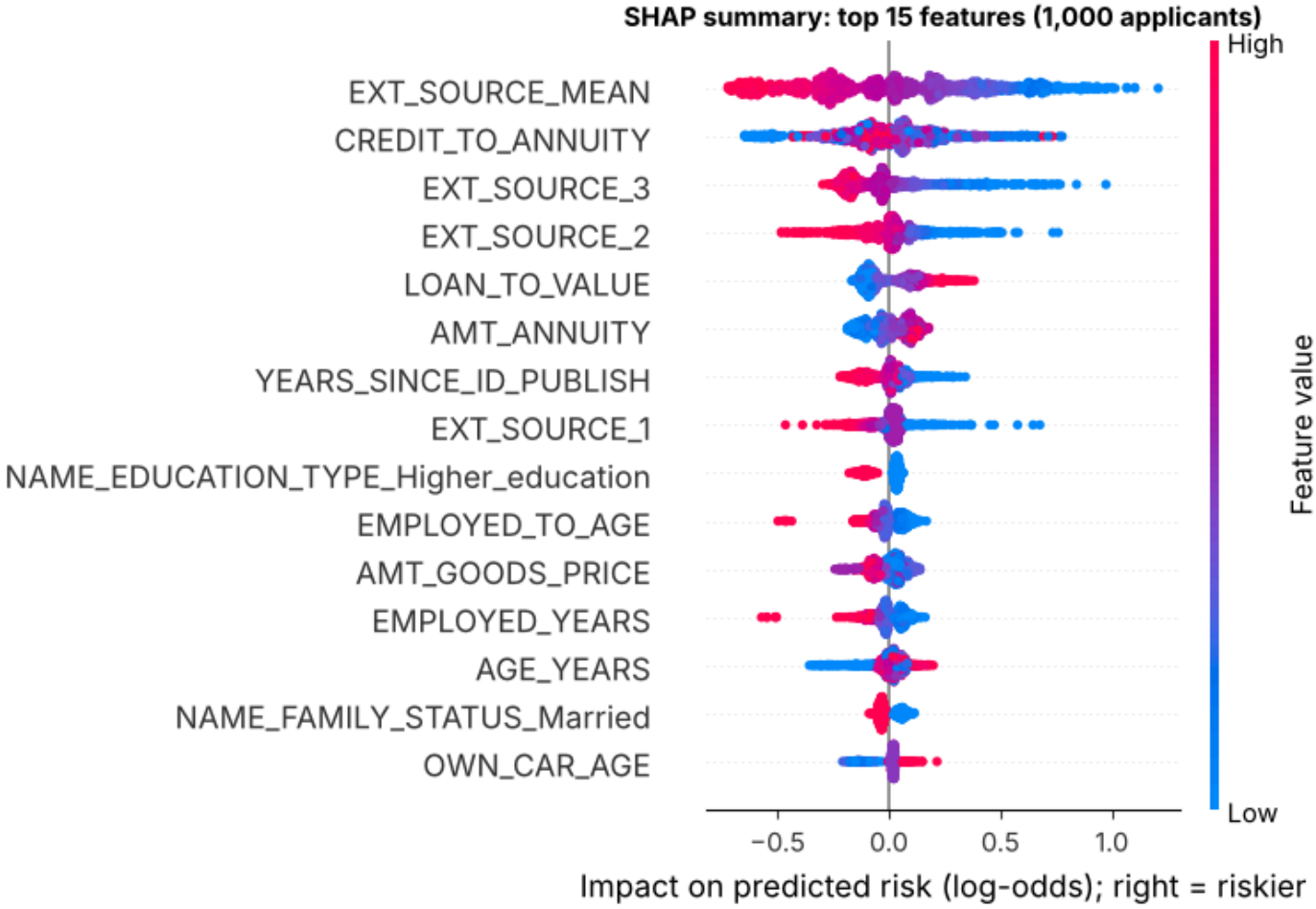
Best threshold under different cost assumptions:

	Margin 5%	Margin 8%	Margin 12%
LGD 45%	9.5% (approve 73%)	14.5% (approve 85%)	21.8% (approve 93%)
LGD 70%	6.5% (approve 60%)	9.8% (approve 74%)	13.2% (approve 83%)
LGD 85%	5.0% (approve 50%)	8.5% (approve 70%)	12.0% (approve 80%)

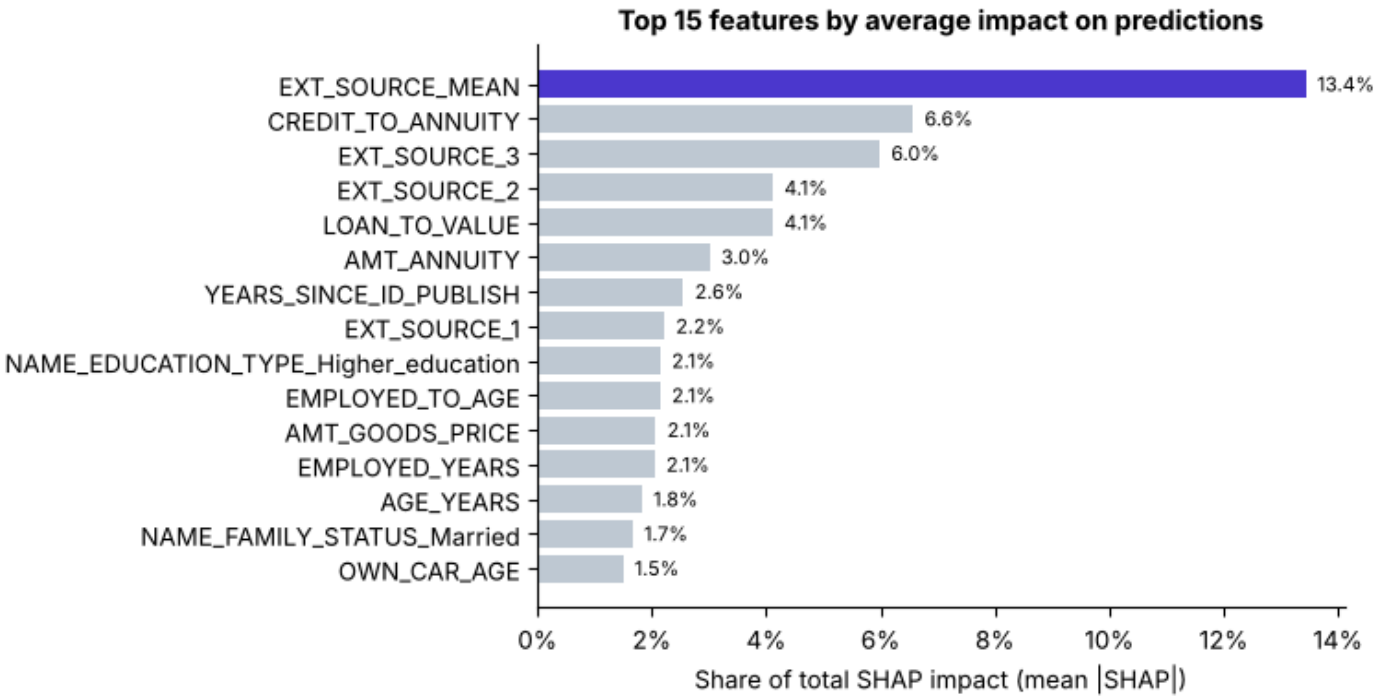
Reading the sensitivity. Lower thresholds cut defaults further but decline many good borrowers, and profit falls; higher thresholds approve more people but let costly defaults through. The best cutoff moves from about 5% to 22% as the cost assumptions change, so **the single most valuable next step is replacing these published figures with Home Credit's own margin and recovery data.** In every scenario the model beats approving everyone; the gain is smallest when margins are high and losses are low.

5 Explainability (SHAP)

SHAP breaks each prediction into contributions from each feature: how much that feature pushed this applicant's risk up or down compared with the average applicant. The summary below uses a random sample of 1,000 applicants.



Each dot is one applicant. Red dots are high values of the feature, blue are low; dots to the right raised that applicant's risk.



Top predictive features in business terms

1. **External credit scores** (`EXT_SOURCE_MEAN`, `_2`, `_3`, `_1`): by far the strongest signal. Low scores sharply raise risk; a missing score is itself a warning sign.
2. **Loan structure** (`CREDIT_TO_ANNUITY`, `AMT_ANNUITY`): the number and size of installments. The effect is non-linear: very short and very long terms behave differently, so it cannot be summarized as "longer is riskier".
3. **Loan amount vs. goods price** (`LOAN_TO_VALUE`, `AMT_GOODS_PRICE`): borrowing well above the price of the item financed raises risk.
4. **Stability signals** (`YEARS_SINCE_ID_PUBLISH`, `EMPLOYED_YEARS`, `EMPLOYED_TO_AGE`): recently issued identity documents and short job tenure raise risk.
5. **Personal characteristics** (`NAME_EDUCATION_TYPE`, `NAME_FAMILY_STATUS`, age): higher education and being married lower predicted risk. These carry legal and fairness implications (Sections 6 and 7).

6 Adverse action mapping

U.S. law (ECOA / Regulation B) requires that declined applicants receive the specific principal reasons for the decision. The table translates each top model feature into a plain-language reason, used only when that feature pushes the applicant's risk **up**.

	Feature	Raises risk when	Adverse action reason given to applicant	Status
1	EXT_SOURCE_MEAN	low	Limited or weak credit history with external credit agencies	Usable
2	CREDIT_TO_ANNUITY	pattern	Requested repayment term	Usable
3	EXT_SOURCE_3	low	Limited or weak credit history with external credit agencies	Usable

	Feature	Raises risk when	Adverse action reason given to applicant	Status
4	EXT_SOURCE_2	low	Limited or weak credit history with external credit agencies	Usable
5	LOAN_TO_VALUE	high	Loan amount is high relative to the price of the goods financed	Usable
6	AMT_ANNUITY	high	Installment amount is high	Usable
7	YEARS_SINCE_ID_PUBLISH	low	Identity document issued or changed recently	Usable
8	EXT_SOURCE_1	low	Limited or weak credit history with external credit agencies	Usable
9	NAME_EDUCATION_TYPE_Higher_education	no	Education level (requires fair lending review)	**Prohibited / review**
10	EMPLOYED_TO_AGE	low	Short employment history	Usable
11	AMT_GOODS_PRICE	low	Loan amount is high relative to the price of the goods financed	Usable
12	EMPLOYED_YEARS	low	Length of employment	Usable
13	AGE_YEARS	pattern	Age (permitted only if applicants 62+ are not disadvantaged: legal review)	**Prohibited / review**
14	NAME_FAMILY_STATUS_Married	no	Marital status (prohibited under ECOA: remove)	**Prohibited / review**
15	OWN_CAR_AGE	high	Age of vehicle owned	Usable

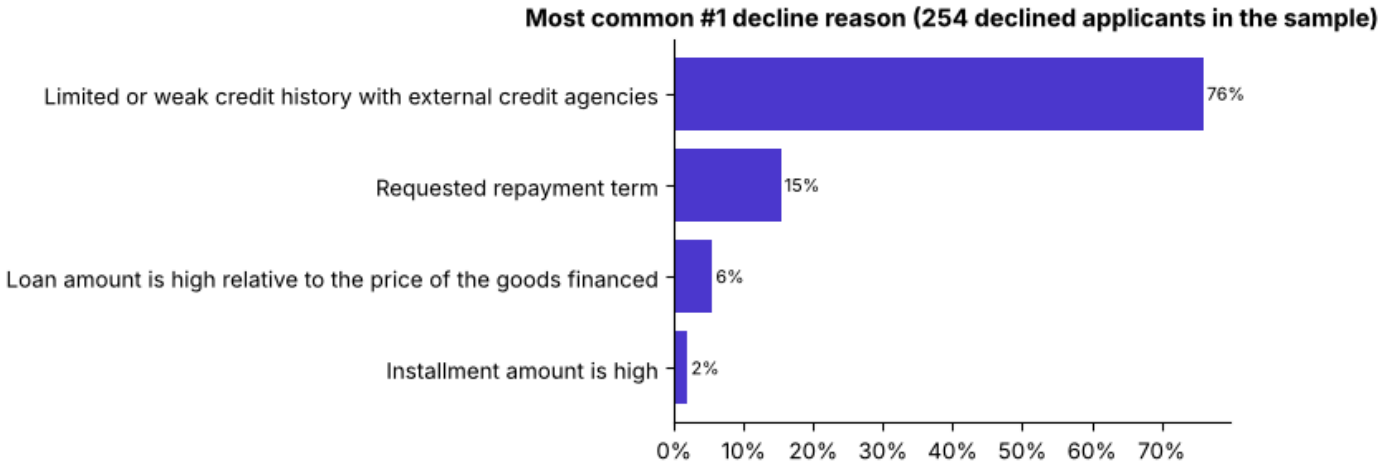
Example: generating reasons for one declined applicant. For each declined applicant, the reasons are the features with the largest risk-raising SHAP values, translated with the table above and de-duplicated.

Predicted default probability: **16.2%** (threshold 10%) → **Decline**

	Risk contribution (log-odds)	Reason
LOAN_TO_VALUE	0.313	Loan amount is high relative to the price of the goods financed
CREDIT_TO_ANNUITY	0.211	Requested repayment term
AMT_ANNUITY	0.137	Installment amount is high
EXT_SOURCE_MEAN	0.116	Limited or weak credit history with external credit agencies
AMT_GOODS_PRICE	0.093	Loan amount is high relative to the price of the goods financed
EXT_SOURCE_3	0.079	Limited or weak credit history with external credit agencies

Principal reasons stated on the adverse action notice:

1. Loan amount is high relative to the price of the goods financed
2. Requested repayment term
3. Installment amount is high
4. Limited or weak credit history with external credit agencies



Limited external credit history is the dominant decline reason. That is accurate, but it means thin-file applicants, the customers Home Credit most wants to serve, are the ones most often declined, which reinforces the case for adding alternative data (Section 8).

7 Fairness analysis

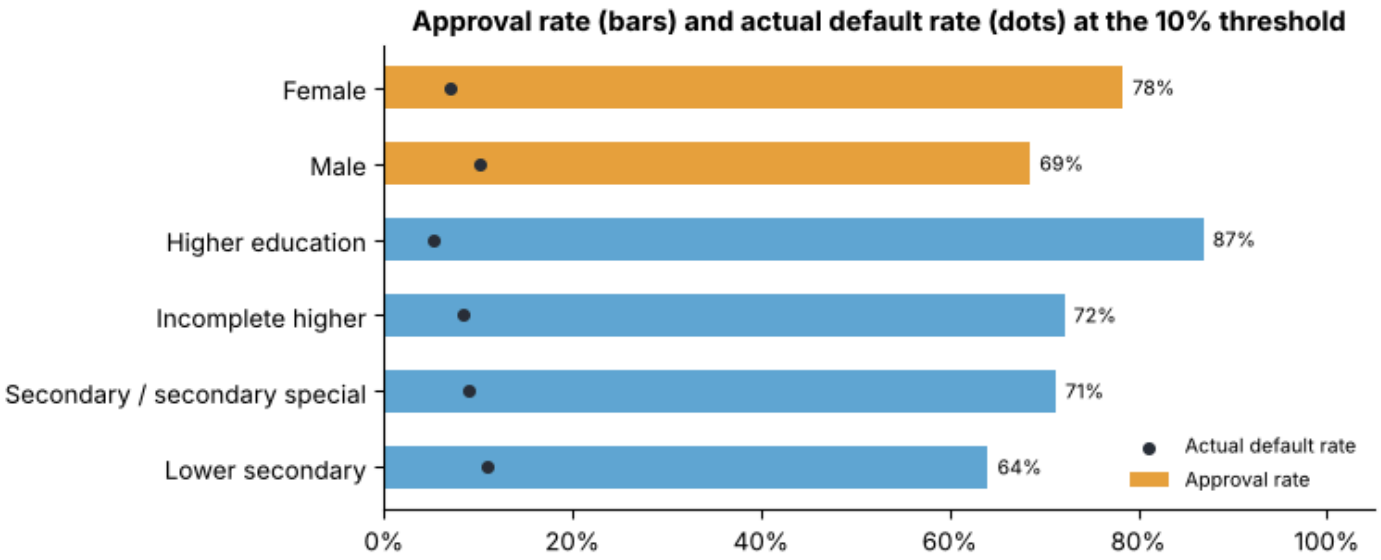
Approval rates at the 10% threshold, compared with each group's actual default rate. The **adverse impact ratio** divides each group's approval rate by the highest group's; values below 0.80 (the "four-fifths rule" used in U.S. disparate impact screening) warrant review.

By gender (gender is not a model input):

	Applicants	Approval rate	Actual default rate	Adverse impact ratio	Approval rate, good borrowers only
Female	202,448	78.4%	7.0%	1.00	81.0%
Male	105,059	68.5%	10.1%	0.87	72.4%

By education level (education is a model input):

	Applicants	Approval rate	Actual default rate	Adverse impact ratio	Approval rate, good borrowers only
Higher education	74,863	86.9%	5.4%	1.00	88.6%
Incomplete higher	10,277	72.2%	8.5%	0.83	75.2%
Secondary / secondary special	218,391	71.3%	8.9%	0.82	74.7%
Lower secondary	3,816	64.1%	10.9%	**0.74**	67.9%



Disparities and implications

- **Gender.** Men are approved less often than women (adverse impact ratio 0.87), even though gender is not an input. Men in this data also default more often, so the gap largely tracks repayment, but the model may also be picking up gender through proxies (occupation, car ownership, external scores). Even among applicants who repaid, men are approved less often, which is the pattern regulators examine.
- **Education.** The gaps are larger. Lower secondary applicants have the lowest approval rate (ratio 0.74). Education is not a protected class under ECOA, but it correlates with income, age and national origin, so using it directly invites disparate impact claims, and it is a weak basis for an adverse action reason.
- **What to do.** (1) Remove marital status, which is prohibited; (2) test a version without education and region and measure the accuracy cost; (3) run a formal disparate impact review (including age 62+) with compliance before deployment; (4) monitor approval and default rates by group every month after launch.

Cost of the compliance fix. Retraining with the same settings but without marital status gives a cross-validated AUC of **0.7687** versus 0.7688 with it, a change of -0.0001. The prohibited variable can be removed at essentially no loss in accuracy.

8 Limitations and risks

Where the model might fail

- **Thin-file applicants.** AUC is 0.748 for the 13.5% of applicants with no bureau inquiry record versus 0.771 for everyone else, and the top decline reason is missing external credit history: the model is least certain about exactly the customers the business wants to grow.
- **Population drift.** The data are a historical snapshot. Changes in the economy, product mix, marketing channels or competitors' policies will shift who applies, and the score will degrade without monitoring and retraining.
- **Selection bias.** The model learned only from loans that were approved in the past. It has never seen outcomes for people Home Credit declined, so its estimates for unusual profiles are extrapolations.

- **Dependence on external scores.** The largest share of the model's signal comes from third-party scores whose methodology Home Credit does not control; if a provider changes its scoring, the model must be revalidated.

What data is missing

- Credit history tables (`bureau.csv`, `previous_application.csv`, installment payments) were not used in this version.
- Verified income and real loan economics: margin, recoveries and collection costs are taken from published sources (Section 4), not Home Credit's books.
- Alternative data that could help thin-file applicants (utility, telecom or transaction history).

Deployment risks

- **Legal and fairness.** Marital status must be removed; age, region and education need compliance review; adverse action reasons must be validated with legal counsel.
- **Cutoff risk.** The profit-optimal threshold moves between about 5% and 22% under plausible cost assumptions; a wrong cutoff either loses money or declines too many good customers.
- **Automation bias.** Loan officers may defer to the score; borderline cases should be reviewed by a person with the reasons visible.
- **Monitoring plan.** Track AUC and calibration monthly on matured loans, approval and default rates by group, and the population stability of key inputs; retrain if AUC falls below 0.74 or calibration drifts.

References

1. Home Credit B.V. (2019). *FY2018 and Q1 2019 Results Announcement* (net interest margin 15.8%, cost-to-income 42.2%, cost of risk 9.4%). https://www.homecredit.net/wp-content/uploads/2022/11/190628-HCBV-FY18_Q1_2019-Results-Announcement.pdf
2. Basel Committee on Banking Supervision (2019). *Basel Framework, CRE32: IRB approach, risk components for each asset class*, paragraph 32.5 (45% LGD for senior unsecured claims). https://www.bis.org/committees/bcbs/basel-framework/81364/chapter.pdf
3. De Almeida Filho, A. T., Mues, C., & Thomas, L. C. (2010). Optimizing the collections process in consumer credit. *Production and Operations Management*, 19(6), 698–708. Working paper: https://eprints.soton.ac.uk/71322/1/CORMSIS-09-07.pdf
4. Mitchell, M. et al. (2019). Model Cards for Model Reporting. *FAT* Conference*. https://arxiv.org/abs/1810.03993
5. Consumer Financial Protection Bureau. *Regulation B (Equal Credit Opportunity Act), 12 CFR 1002.9: notifications and statement of specific reasons*. https://www.consumerfinance.gov/rules-policy/regulations/1002/9/
6. Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *NeurIPS*. https://arxiv.org/abs/1705.07874

Use of AI

I used Claude (Anthropic) to draft this model card, run the SHAP and threshold calculations, and locate published cost figures. I chose the cost assumptions, checked each source, and made the judgment calls on fairness and compliance.